

Installation of gcc and creation of main.c -> output into main and run

```
37 sudo apt install gcc
38 clear
39 gcc main.c -o main
40 sudo gcc main.c -o main
41 clear
42 ls
43 ./main
44 history
ashen@Ashen:/mnt/c/Users/dznel/Desktop/Ynov/compiledLanguages$
```

Output

```
ashen@Ashen:/mnt/c/Users/dznel/Desktop/Ynov/compiledLanguages$ ./main
(lire avec accent Mexican): Bonjourno compilation !ashen@Ashen:/mnt/c/Users/dznel/Desktop/Ynov/compiledLanguages$ history
```

Atelier 1 :

- gcc verification

```
ashen@Ashen:/mnt/c/Users/dznel/Desktop/Ynov/compiledLanguages$ gcc --version
gcc (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0
Copyright (C) 2023 Free Software Foundation, Inc.
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
```

- Name / Age on two lines

```
ashen@Ashen:/mnt/c/Users/dznel/Desktop/Ynov/compiledLanguages$ ./main
Je m'appelle Luka
et j'ai 21 ans
```

- Verify with other than 0

```
ashen@Ashen:/mnt/c/Users/dznel/Desktop/Ynov/compiledLanguages$ echo $?
2
ashen@Ashen:/mnt/c/Users/dznel/Desktop/Ynov/compiledLanguages$
```

- error message

```
ashen@Ashen:/mnt/c/Users/dznel/Desktop/Ynov/compiledLanguages$ sudo gcc main.c -o main
main.c: In function 'main':
main.c:6:9: error: expected ',' or ';' before 'printf'
   6 |         printf("Je m'appelle %s\net j'ai %d ans\n", prenom, age);
     |         ~~~~~^
ashen@Ashen:/mnt/c/Users/dznel/Desktop/Ynov/compiledLanguages$
```

TP 1 :

1.1 Tableau Comparative :

Critere	Langage compile (C, C++, Rust)	Langage interprete (Python, JS)
Quand la traduction se fait-elle ?	During the compilation	During the runtime
Vitesse d'execution	Faster	Slower
Portabilite du binaire	Not portable (doesn't work well with different CPU's)	More portable than compiled
Detection des erreurs	During compilation	During runtime
Besoin d'un runtime installe ?	Yes	Yes
Exemples de langages	Golang	Ruby

1.2 Questions de reflexion :

- 1. C is faster because it compiles the code once and then run's it while python needs to compile SourceCode at Runtime each time
- 2. No a windows compiled code won't run on Linux unless specifically coding it for Linux also. This is because the language uses OS-Specific API's to compile. There are alternatives like Wine to run .exe files on Linux though but also depends on coded program.
- 3. No the interpreted languages themselves aren't made with C but the Interpreters made for Python for example are made via CPython which is one of the runtime compilers based on C but this can also change by adding another one like PyPy and or a JIT interpreter. But this means that Compiled languages like C are cornerstone of every other languages Interpreted or not. CPython is for example based on C because of it's speed.
- 4. For example games that are compiled can be compiled into an .exe file which would run faster than launching a python file each time.
- 5. Example 1 : Jets, they need precise CPU control and as quick as possible real time execution speed | Example 2 : Video games that have high graphical dependency need to execute multiple frames per second and need fast execution (engines made in C++)

2.1 Done Above

2.2 Done Above

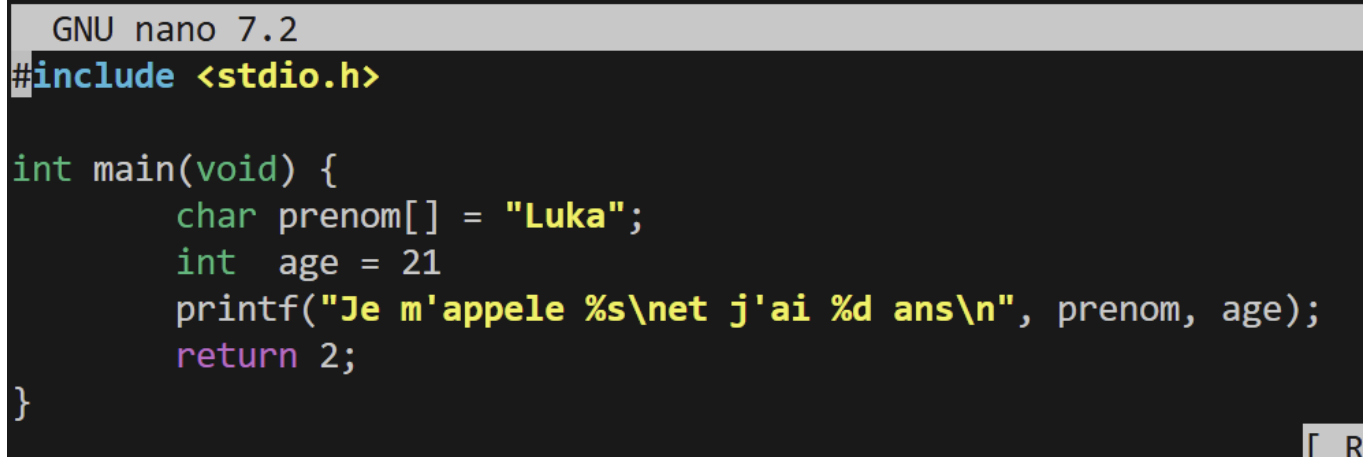
3.1 Done Above

3.2 Done Above

3.3 Done Above

3.4 A : Done Above

3.4 B :



```
GNU nano 7.2
#include <stdio.h>

int main(void) {
    char prenom[] = "Luka";
    int age = 21
    printf("Je m'appelle %s\net j'ai %d ans\n", prenom, age);
    return 2;
}
```

4.1 Done Above

4.2 Done Above

4.3 :

Program 1 errored :

```
#include
int main(void) {
    printf("Bonjour le monde!")
    return 0;
}
```

Program 1 Corrected :

```
#include <stdio.h>
int main(void) {
    printf("Bonjour le monde!");
}
```

```
    return 0;
}
```

Program 2 Errored :

```
#include
int main(void) {
    int nombre = 42
    float pi = 3.14;
    printf("Nombre : %f, Pi : %d\n", nombre, pi);
    return 0;
}
```

Program 2 Corrected :

```
#include <stdio.h>
int main(void) {
    int nombre = 42;
    float pi = 3.14;
    printf("Nombre : %d, Pi : %f\n", nombre, pi);
    return 0;
}
```

Program 3 Errored :

```
ashen@Ashen:/mnt/c/Users/dznel/Desktop/Ynov/compiledLanguages$ ./test1
Floating point exception (core dumped)
```

```
#include <stdio.h>
int main(void) {
    int a = 10;
    int b = 0;
    int resultat = a / b;
    printf("Resultat : %d\n", resultat);
    return 0;
}
```

Program 3 Corrected : Can't divide by 0

```
ashen@Ashen:/mnt/c/Users/dznel/Desktop/Ynov/compiledLanguages$ ./test1
Resultat : 10
```

```
#include <stdio.h>
int main(void) {
    int a = 10;
    int b = 1;
    int resultat = a / b;
    printf("Resultat : %d\n", resultat);
    return 0;
}
```

Program 4 Errored :

```
#include <stdio.h>
int main(void) {
    int score = 85;
    if (score = 100) {
        printf("Score parfait !\n");
    }
    return 0;
}
```

Program 4 Corrected :

```
ashen@Ashen:/mnt/c/Users/dznel/Desktop/Ynov/compiledLanguages$ ./test2
Score bellow 100
```

```
#include <stdio.h>
int main(void) {
    int score = 85;
    if (score == 100) {
        printf("Score parfait !\n");
    } else {
        printf("Score bellow 100\n");
    }
    return 0;
}
```

4.4

```
ashen@Ashen:/mnt/c/Users/dznel/Desktop/Ynov/compiledLanguages$ sudo gcc -Wall -Wextra test2.c -o test3
test2.c: In function 'main':
test2.c:4:13: warning: suggest parentheses around assignment used as truth value [-Wparentheses]
  4 |         if (score = 100) {
    |             ^~~~~~
```

Basically running with `-Wall -Wextra` warns you about potential errors or incoherent logic that might be inside your code. So it's good to compile like this to preview errors.

5 :

Errored :

```
ashen@Ashen:/mnt/c/Users/dznel/Desktop/Ynov/compiledLanguages$ sudo gcc -Wall -Wextra fiche.c -o fiche
fiche.c: In function 'main':
fiche.c:7:24: warning: initialization of 'char' from 'char *' makes integer from pointer without a cast [-Wint-conversion]
   7 |         char initial = "L";
     |         ^~~~~~
```

Corrected :

```
ashen@Ashen:/mnt/c/Users/dznel/Desktop/Ynov/compiledLanguages$ sudo gcc -Wall -Wextra fiche.c -o fiche
ashen@Ashen:/mnt/c/Users/dznel/Desktop/Ynov/compiledLanguages$ ./fiche
+-----+
|      FICHE D'IDENTITE      |
+-----+
| Prenom  : Luka             |
| Age     : 21                |
| Taille  : 1.67 m           |
| Initiale: L                 |
+-----+
| Code de sortie : 0 (succes) |
+-----+
```